From Nothing to Coulomb's Law

Electricity — lesson notes · discussion of 12 July 2026 · Dad's notes, corrected and extended

The chain of ideas

1. **Everything is made of particles.**
   Matter is made of atoms, and atoms are made of protons, neutrons and electrons.

2. **Electric charge is a property of a particle.**
   Just as a particle has mass, it can carry charge. Charge is measured in coulombs (C).

3. **Particles can be positive, negative, or neutral.**
   The proton is positive, the electron is negative, the neutron is neutral.

4. **Charge is quantised: it comes in steps of the elementary charge e.**
   Every observed charge is a whole-number multiple of e. The proton carries exactly $+e$ and the electron exactly $-e$. (Fine print: quarks carry $\pm e/3$ and $\pm 2e/3$, but they are never seen alone.)

5. **Charged particles interact: they attract or repel each other.**
   Neutral particles feel no electric force.

6. **Like charges repel; unlike charges attract.**
   + and + push apart, − and − push apart, + and − pull together.

7. **The net charge of a body counts its imbalance of particles: $Q = (N_+ - N_-)\, e$.**
   N_+ = number of positive charges (protons), N_- = number of negative ones (electrons). Most bodies have $N_+ \approx N_-$, so they are neutral.

8. **Charge is conserved.**
   Charge is never created or destroyed — it only moves from one body to another (e.g. rubbing a balloon moves electrons onto it).

9. **Coulomb's law tells us how strong the force is.**
   See the box below — the climax of the chain.

Coulomb's law

$$F \;=\; k\,\frac{|q_1|\,|q_2|}{r^2} \qquad \text{with} \qquad k \;=\; \frac{1}{4\pi\varepsilon_0} \;\approx\; 8.99\times 10^{9}\ \mathrm{N\,m^2/C^2}$$

- q_1, q_2 — the two charges (in coulombs); r — the distance between them (in metres). Valid for **point charges** (small compared with r).
- The force acts **along the line joining the charges**: like charges are pushed apart, unlike charges pulled together. Both charges feel the **same** magnitude of force, in opposite directions (Newton's third law).
- **Inverse square:** double the distance ⇒ one quarter of the force.

Worked example. Two charges of $1\,\mu$C each, 10 cm apart:

$$F \;=\; 8.99\times 10^{9}\cdot\frac{(10^{-6})^2}{(0.1)^2} \;\approx\; 0.9\ \mathrm{N}$$ — about the weight of a small apple.

Scale check: two charges of 1 C at 1 m would push with 9×10^9 N (≈ the weight of 900 000 tonnes) — that is why real charges are microcoulombs.

Constants worth remembering

Quantity	Symbol	Value	Note
Elementary charge	e	1.602×10^{-19} C	exact: $1.602\,176\,634 \times 10^{-19}$ C
Electron charge	$-e$	-1.602×10^{-19} C	proton: $+e$
Coulomb constant	$k = 1/(4\pi\varepsilon_0)$	8.99×10^{9} N m^2/C^2	remember as $\approx 9 \times 10^9$
Vacuum permittivity	ε_0	8.854×10^{-12} C^2/(N m^2)	the same thing as F/m
Charges in 1 coulomb	$1\,\text{C}/e$	6.24×10^{18}	1 C is a *huge* charge
Electron mass	m_e	9.11×10^{-31} kg	proton is ≈1836× heavier

Where to read (from our archive)

- **Siyavula, Physical Sciences Gr 10** — `docs/textbook/siyavula-physical-sciences/Gr10_PhysicalSciences_Learner_Eng.pdf`
  ch 16 *Electrostatics* — §16.2 Two kinds of charge (PDF p. 276), §16.3 Conservation of charge (PDF p. 282), §16.4 Quantisation of charge (PDF p. 284). Closest match to these notes.

- **CK-12, Physical Science Workbook** — `docs/textbook/ck12-physical-science/CK-12-Physical-Science-N pdf`
  lesson 23.1 *Electric Charge* (printed p. 276 ≈ PDF p. 283) — practice worksheets; answers in ch 26.23 (printed p. 417).

- **NCERT, Class 9 Science** — `docs/textbook/ncert-class9-physics/other-chapters/ch08-journey-inside-t pdf`
  §8.2–8.6 (PDF pp. 2–12) — protons & electrons, atomic number: why atoms are neutral and where the charges sit.

- **Cambridge Lower Secondary, Stage 7** — `docs/textbook/cambridge/Cambridge_Lower_Secondary_Science_2ed_7_Learners_book.pdf`
  Unit 9, §9.1 *Flow of electricity* (≈ p. 287) — moving charge = current: the next step after these notes.

Full quantitative treatment: **Siyavula Grade 11** (free, openly licensed) — easy to add to the archive.